ECS 408/608
Assignment 3
Interprocess Communication
Solutions

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For code repository of this assignment: **Operating Systems Assignments**

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1. Problem 1

Problem statement

Write C program using IPC-Shared Memory that performs following two task using producer-consumer paradigm:

(a) Producer (producer.c): Ask the user for an integer and string and place them in shared memory.

(b) Consumer (consumer.c): Print the string input in the producer and check if the input number is even or not.

Answers

1.1 Code:

Shared:

```
#ifndef SHARED_H
#define SHARED_H

struct shared_data {
    int number;
    char text[100];
};

#endif
```

Producer:

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>
#include "shared.h"

int main() {

    key_t key = 1234;

```

```
    int shmid = shmget(key, sizeof(struct shared_data), 0666 | IPC_CREAT);

    struct shared_data *data = (struct shared_data*) shmat(shmid, NULL, 0);

    printf("Enter an integer: ");
    scanf("%d", &data->number);

    printf("Enter a string: ");
    scanf("%s", data->text);

    printf("Data written to shared memory\n");

    shmdt(data);

    return 0;
}
```

Consumer:

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include "shared.h"

int main() {

    key_t key = 1234;

    int shmid = shmget(key, sizeof(struct shared_data), 0666);

    struct shared_data *data = (struct shared_data*) shmat(shmid, NULL, 0);

    printf("String from producer: %s\n", data->text);

    if(data->number % 2 == 0)
        printf("Number %d is EVEN\n", data->number);
    else
        printf("Number %d is ODD\n", data->number);

    shmdt(data);

    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}
```

1.2 Output:

a)

```
Enter an integer: 5
Enter a string: abc
```

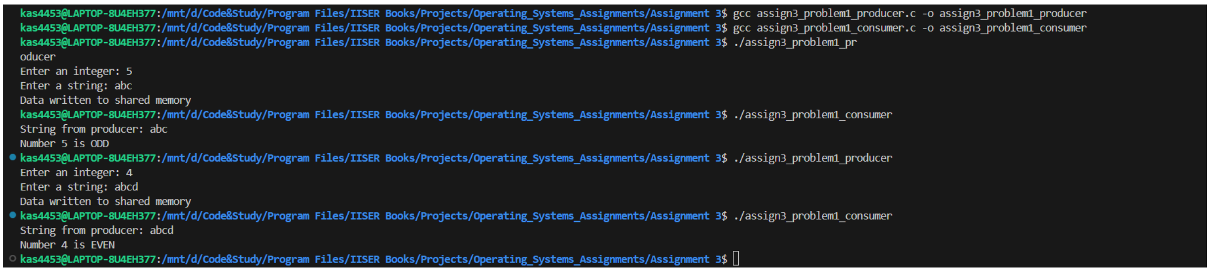
```
Data written to shared memory

String from producer: abc
Number 5 is ODD
```

b)

```
Enter an integer: 4
Enter a string: abcd
Data written to shared memory

String from producer: abcd
Number 4 is EVEN
```

Output Screenshots

Terminal output



```
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem1_producer.c -o assign3_problem1_producer
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem1_consumer.c -o assign3_problem1_consumer
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem1_pr
oducer
Enter an integer: 5
Enter a string: abc
Data written to shared memory
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem1_consumer
String from producer: abc
Number 5 is ODD
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem1_producer
Enter an integer: 4
Enter a string: abcd
Data written to shared memory
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem1_consumer
String from producer: abcd
Number 4 is EVEN
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$
```

2. Problem 2

Problem statement

Write a C program using IPC-Shared Memory that perform the following task using producer-consumer paradigm:

(a) Producer (producer.c): Generate 100 random numbers and place them in shared memory.

(b) Consumer (consumer.c): Read all the 100 numbers and generate the cumulative sum and exit.

Answers

2.1 Code:

Producer:

```c
#include <stdio.h>
#include <stdlib.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <time.h>

#define SIZE 100

int main() {

    key_t key = 5678;

    int shmid = shmget(key, SIZE * sizeof(int), 0666 | IPC_CREAT);

    int *numbers = (int*) shmat(shmid, NULL, 0);

    srand(time(NULL));

    printf("Generated Numbers:\n");

    for(int i=0;i<SIZE;i++)
    {
        numbers[i] = rand() % 100;
        printf("%d ", numbers[i]);
    }

    printf("\nNumbers written to shared memory\n");

    shmdt(numbers);

    return 0;
}
```

Consumer:

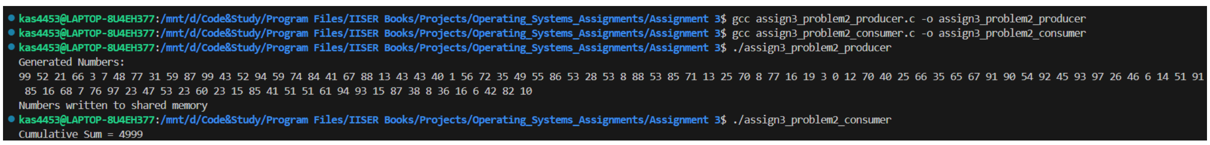
```c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

#define SIZE 100

int main() {

    key_t key = 5678;

    int shmid = shmget(key, SIZE*sizeof(int), 0666);

    int *numbers = (int*) shmat(shmid, NULL, 0);

    int sum = 0;

    for(int i=0;i<SIZE;i++)
        sum += numbers[i];

    printf("Cumulative Sum = %d\n", sum);

    shmdt(numbers);

    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}
```

2.2 Output:

```
Generated Numbers:
99 52 21 66 3 7 48 77 31 59 87 99 43 52 94 59 74 84 41 67 88 13 43 43 40 1 56 72 35 49 55 86 53 28 53 8 88 53 85 71 13 25 70 8 77 16 19 3 0 12 70 40 25 66 35 65 67 91 90 54 92 45 93 97 26 46 6 14 51 91 85 16 68 7 76 97 23 47 53 23 60 23 15 85 41 51 51 61 94 93 15 87 38 8 36 16 6 42 82 10
Numbers written to shared memory

Cumulative Sum = 4999
```

Output Screenshots

Terminal output



3. Problem 3

Problem statement

Write a C program using IPC-Shared Memory that perform the following tasks:

(a) Use shared memory to store an array of numbers.

(b) The parent process writes 10 numbers into the shared memory.

(c) Create 3 consumer processes where: i. Consumer 1 computes the factorial of the first four numbers.

ii. Consumer 2 computes the factorial of the next two numbers. iii. Consumer 3 computes the remaining numbers.

Answers

3.1 Code:

```c
#include <stdio.h>
#include <stdlib.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>
#include <sys/wait.h>

#define SIZE 10

long factorial(int n)
{
    long f = 1;
    for(int i = 1; i <= n; i++)
        f *= i;
    return f;
}

int main()
{
    key_t key = 5678;

    int shmid = shmget(key, SIZE * sizeof(int), 0666 | IPC_CREAT);

    int *arr = (int *)shmat(shmid, NULL, 0);

    printf("Enter 10 numbers:\n");

    for(int i = 0; i < SIZE; i++)
        scanf("%d", &arr[i]);

    /* Consumer 1 */
```

```c
    if(fork() == 0)
    {
        printf("\nConsumer 1 (first 4 numbers)\n");
        for(int i = 0; i < 4; i++)
            printf("Factorial of %d = %ld\n", arr[i], factorial(arr[i]));
        exit(0);
    }

    /* Consumer 2 */
    if(fork() == 0)
    {
        printf("\nConsumer 2 (next 2 numbers)\n");
        for(int i = 4; i < 6; i++)
            printf("Factorial of %d = %ld\n", arr[i], factorial(arr[i]));
        exit(0);
    }

    /* Consumer 3 */
    if(fork() == 0)
    {
        printf("\nConsumer 3 (remaining numbers)\n");
        for(int i = 6; i < 10; i++)
            printf("Factorial of %d = %ld\n", arr[i], factorial(arr[i]));
        exit(0);
    }

    /* wait for children */
    for(int i = 0; i < 3; i++)
        wait(NULL);

    shmdt(arr);
    shmctl(shmid, IPC_RMID, NULL);

    return 0;
}
```

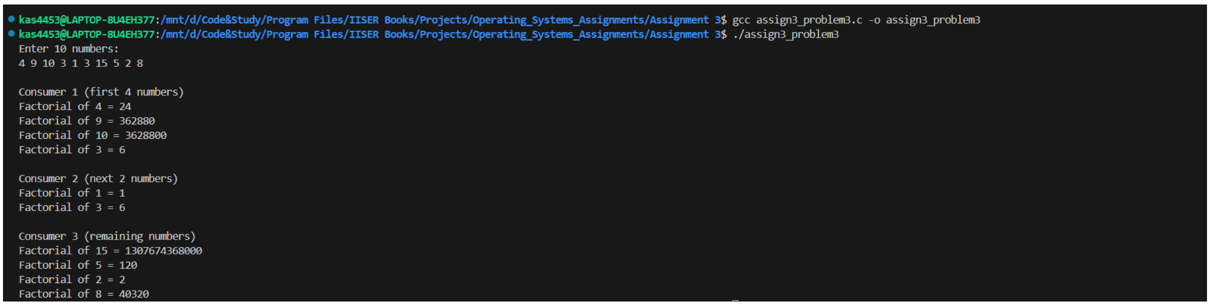
3.2 Output:

```
Enter 10 numbers:
4 9 10 3 1 3 15 5 2 8

Consumer 1 (first 4 numbers)
Factorial of 4 = 24
Factorial of 9 = 362880
Factorial of 10 = 3628800
Factorial of 3 = 6

Consumer 2 (next 2 numbers)
Factorial of 1 = 1
Factorial of 3 = 6
```

```

Consumer 3 (remaining numbers)
Factorial of 15 = 1307674368000
Factorial of 5 = 120
Factorial of 2 = 2
Factorial of 8 = 40320
```

Output Screenshots

Terminal output



```
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem3.c -o assign3_problem3
kas4453@LAPTOP-8U4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem3
Enter 10 numbers:
4 9 10 3 1 3 15 5 2 8

Consumer 1 (first 4 numbers)
Factorial of 4 = 24
Factorial of 9 = 362880
Factorial of 10 = 3628800
Factorial of 3 = 6

Consumer 2 (next 2 numbers)
Factorial of 1 = 1
Factorial of 3 = 6

Consumer 3 (remaining numbers)
Factorial of 15 = 1307674368000
Factorial of 5 = 120
Factorial of 2 = 2
Factorial of 8 = 40320
```